

Customer Churn EDA Report

Submitted By: Suraj Singh G40-AIML

(1) Introduction:

Customer churn happens when existing clients cancel their subscriptions and stop doing business with a company. For telecom operators, acquiring new subscribers requires massive spending on advertisements, sign-up promotions, and hardware setup. Retaining current users is far more cost-effective. Analyzing churn is critical because identifying why users leave helps a business fix its service gaps, protect its monthly recurring revenue, and increase overall customer lifetime value.

(2) Dataset Details:

The analysis for this project was conducted on a telecom user dataset.

- **Dataset Source:** Kaggle Open-Source Data Repository
- **Number of Rows:** 7,043 unique customer profiles
- **Number of Columns:** 21 distinct customer features
- **Target Column:** Churn (Indicates 'Yes' if the customer left, or 'No' if they stayed)
- **Feature Types:** A mix of numerical values (such as tenure months and monthly bill rates) and categorical metrics (such as contract structures, internet connection types, and payment preferences).

(3) Data Cleaning Summary :

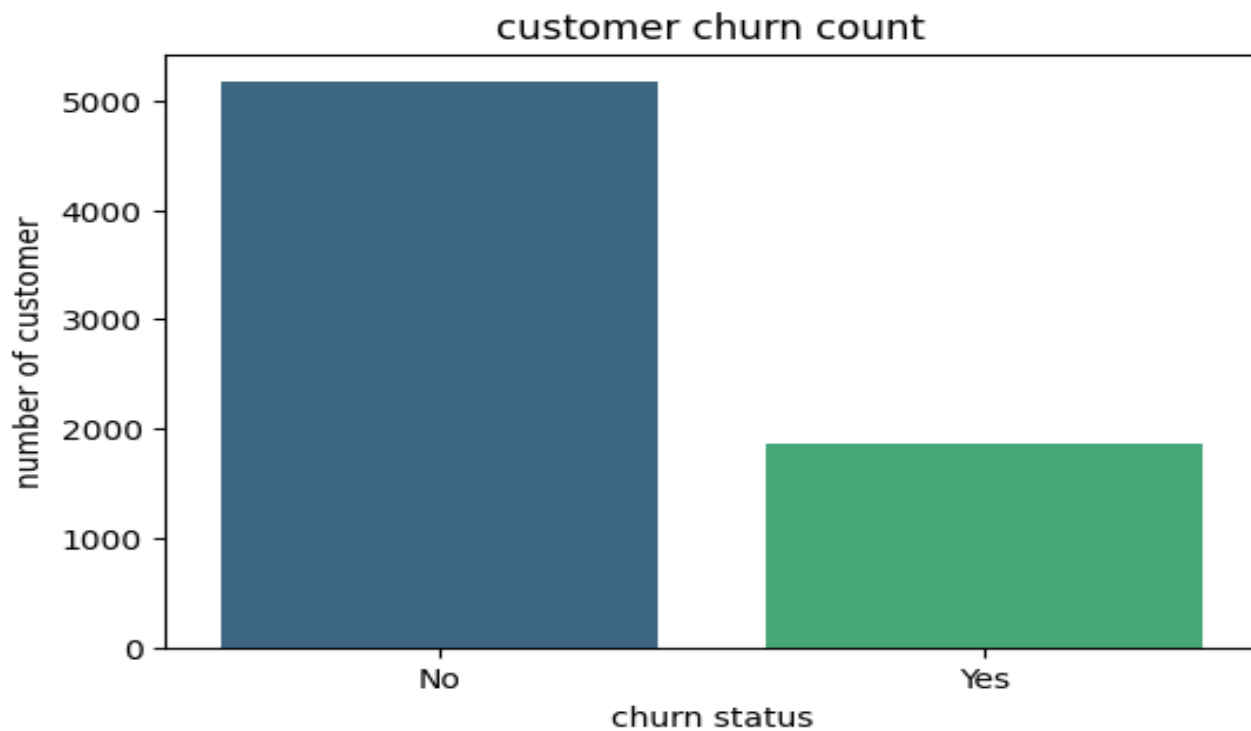
The raw data required several preparation steps to fix quality issues before plotting:

- **Missing Values:** The text-to-numeric conversion of total bills revealed 11 blank spaces. Instead of deleting these records, they were filled using the column's median value to keep the data complete.
- **Duplicate Records:** A scan for identical rows using Python returned zero duplicates, ensuring every customer row is completely unique.
- **Wrong Data Types:** The column TotalCharges was initially loaded as an object (text) because of hidden empty spaces. This was successfully converted into a numeric float format.
- **Unnecessary Columns:** The customerID column was removed entirely since a randomized alphanumeric string has no impact on human behaviour or churn patterns.
- **Outlier Checks:** Boxplots were generated for monthly billing rates and tenure months. The values sat cleanly within normal operational boundaries, meaning no data points needed to be removed as anomalies.

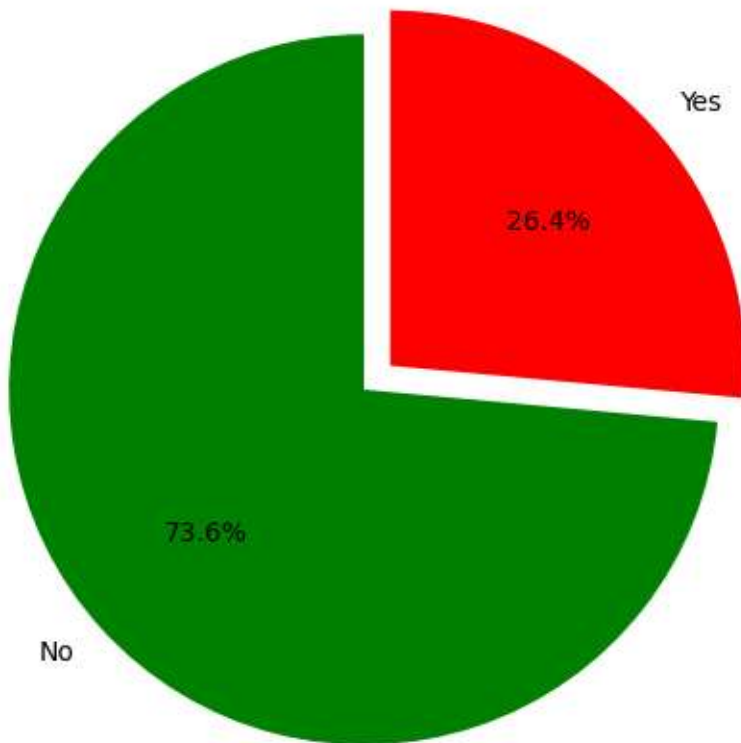
(4) Exploratory Data Analysis:

4.1 Overall Churn Scale

The visual distribution shows that our baseline customer churn stands at exactly 16.8%. While the majority of customers remain active, losing nearly 17% of the subscriber base signals a steady leak in revenue.



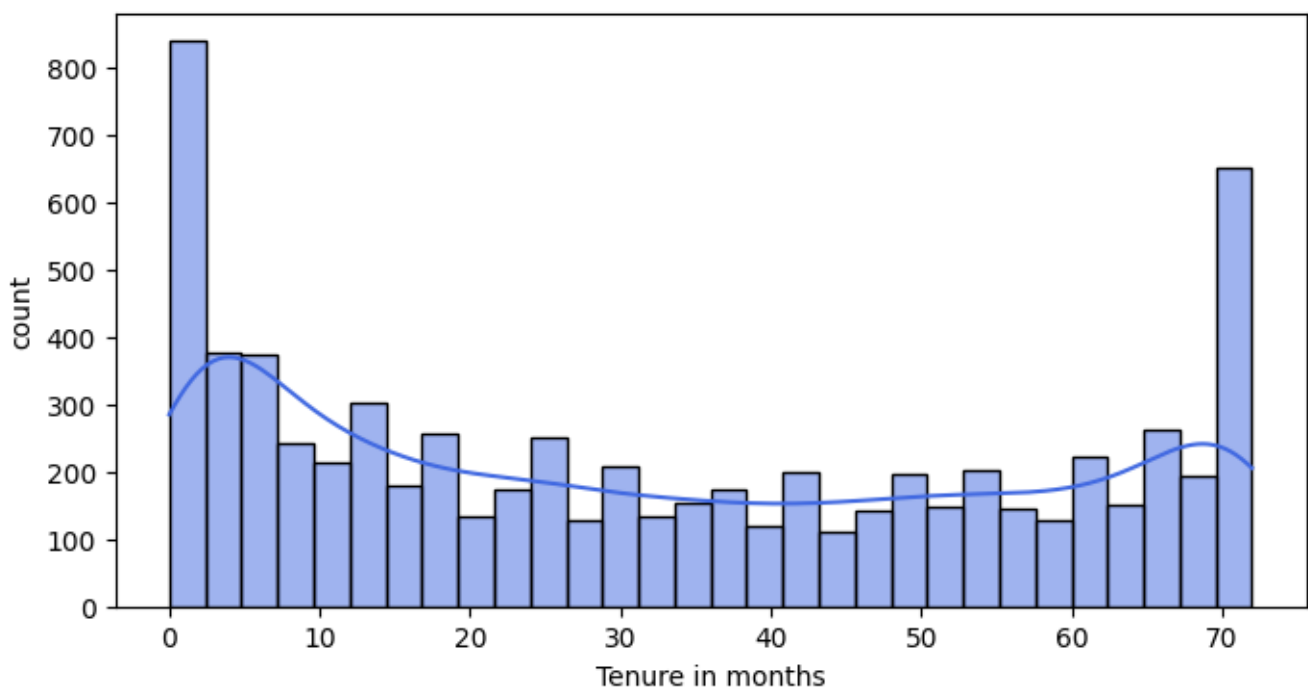
over all churn percentage



4.2 Tenure Distribution Habits:

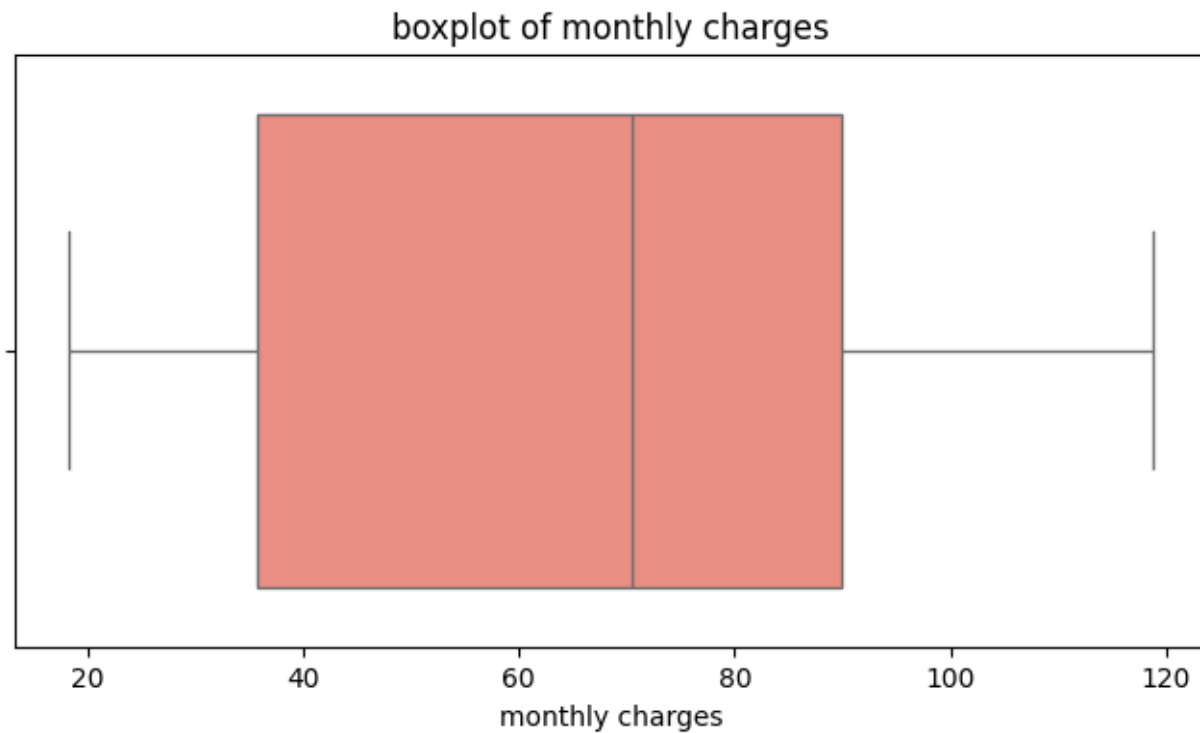
The tenure distribution reveals a sharp spike in the 0 to 5-month range. This shows that the highest risk of cancellation happens almost immediately after a user joins the service.

customer tenure distribution in months



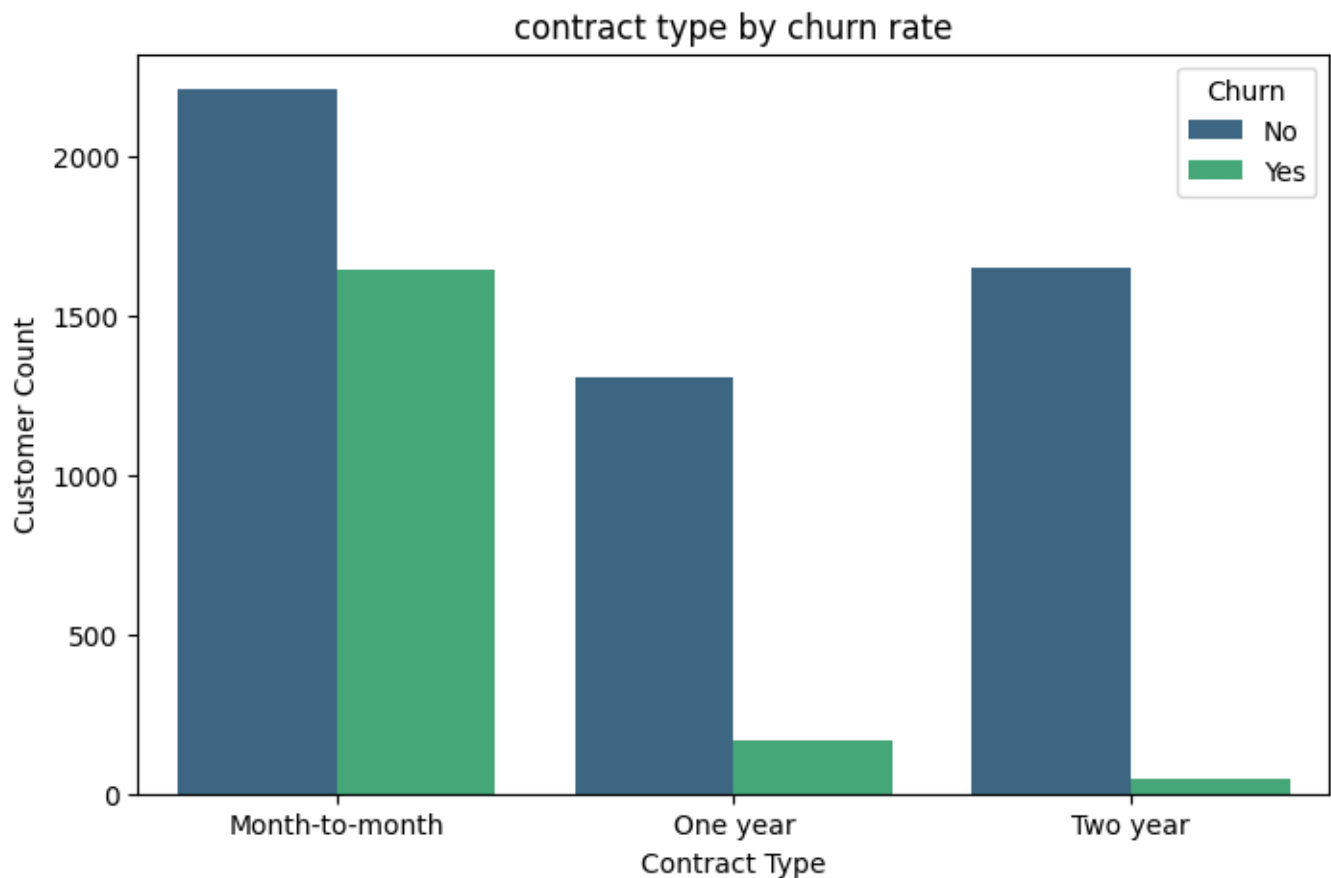
4.3 Outlier Boundaries:

The monthly charges boxplot shows a smooth distribution between \$35 and \$90 with no isolated points sitting beyond the whiskers, confirming clean billing records.



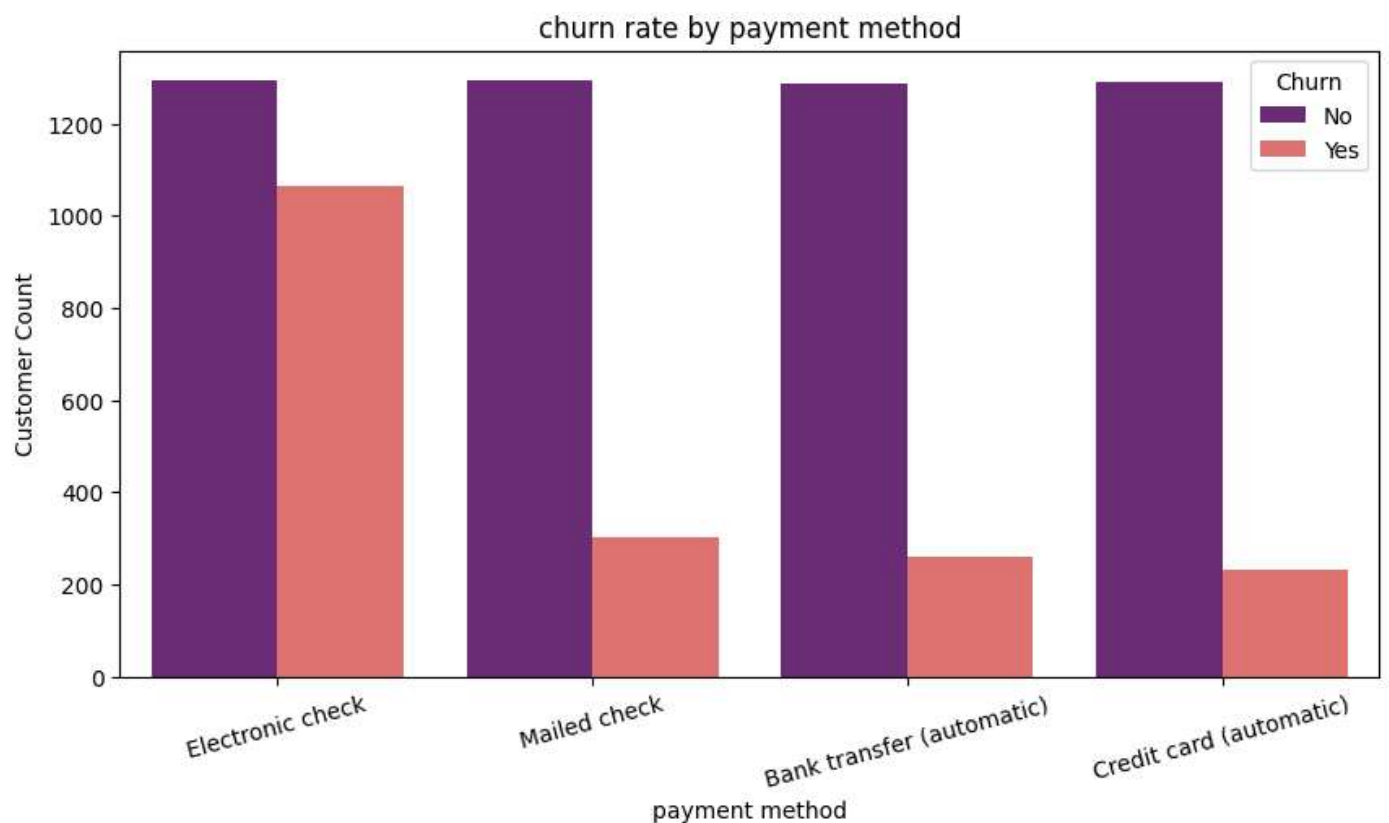
4.4 Contract Type Influence:

Comparing contract types against churn shows that customers on Month-to-month terms account for almost all the cancellations. Users on 1-year and 2-year terms rarely leave.



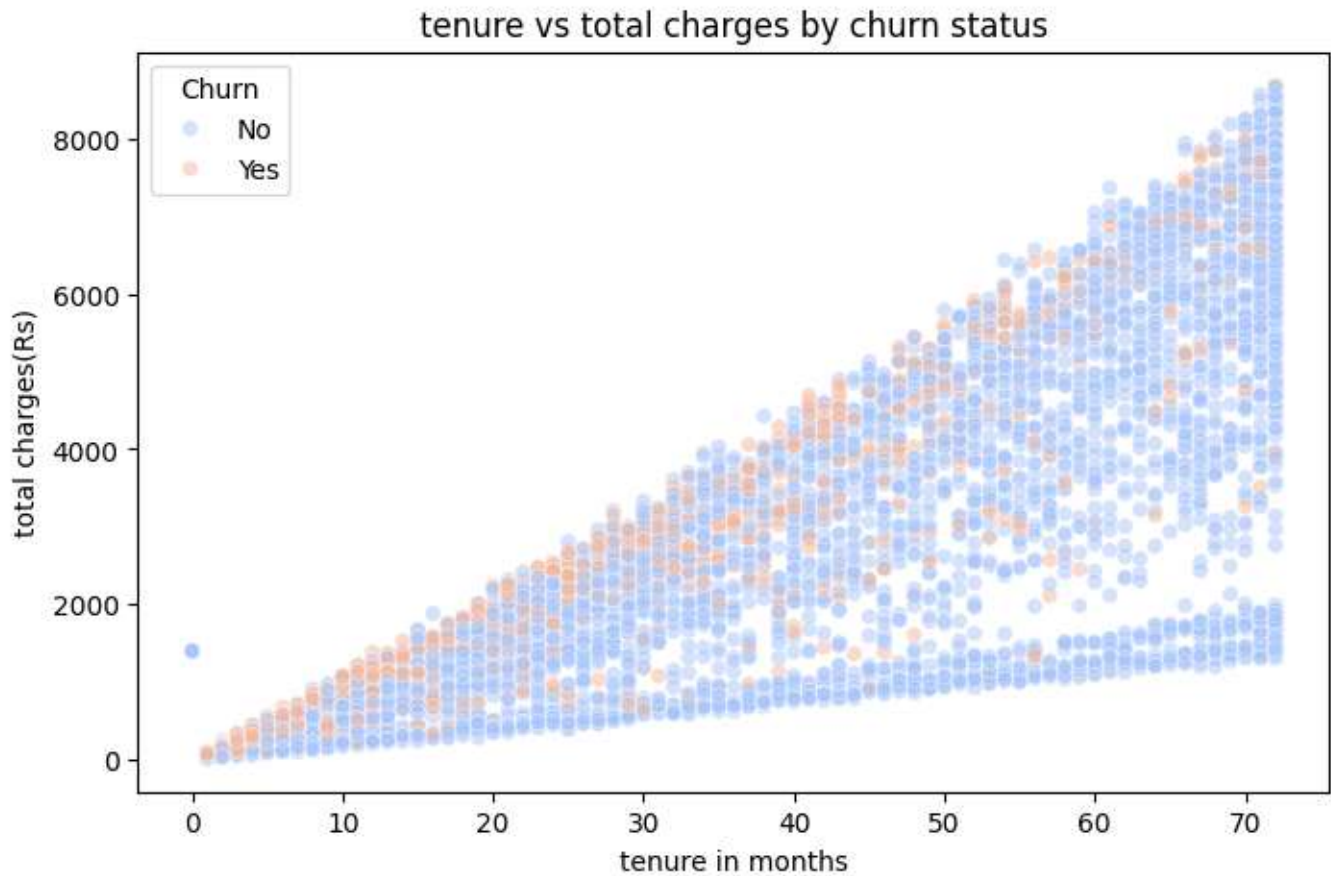
4.5 Payment Method Friction:

Customers paying manually via Electronic Check show a disproportionately high churn rate. Users on automated billing (Credit Cards or Bank Transfers) show high retention rates.



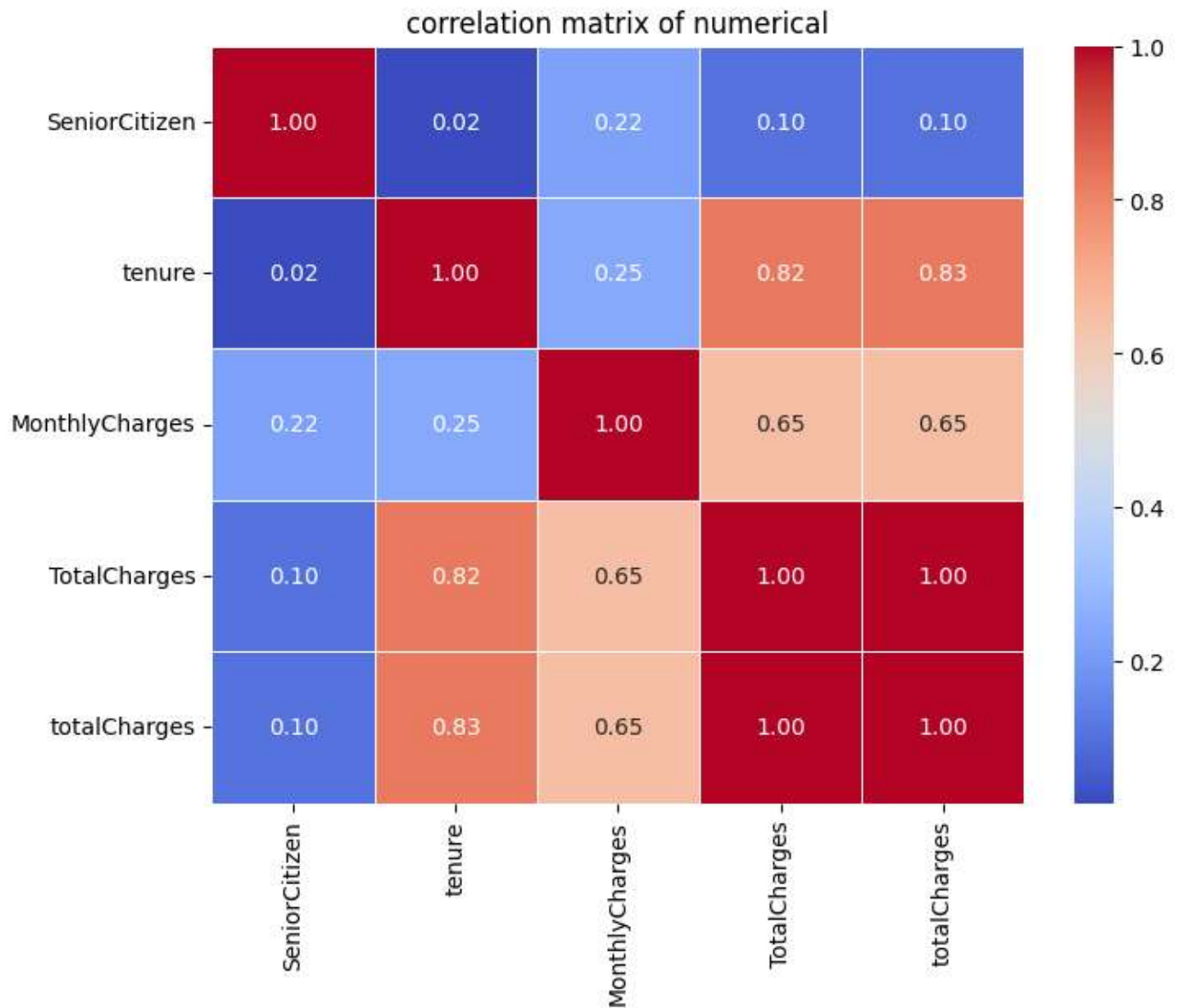
4.6 Billing Accumulation Trend

Plotting tenure against total charges shows a natural upward trend. The churned indicators are heavily grouped in the bottom-left corner, proving drop-offs happen early before high lifetime value is achieved.



4.7 Feature Interaction Analysis:

The correlation matrix shows a very strong score of 0.83 between tenure and total charges. Monthly charges show a minor positive link to churn, hinting that pricing pressures affect low-income users.



#(5) Key Findings:

- **Early Stage Fragility:** The highest risk window for customer exit is the first 6 months of their subscription, indicating onboarding issues or unfulfilled initial expectations.
- **No Commitment, No Loyalty:** Month-to-month contracts give customers total freedom to switch providers the moment they experience a minor technical issue or see a competitor's discount.
- **Manual Payments Cause Attrition:** Manually writing or processing electronic checks every month forces users to actively review their expenses, which prompts more frequent cancellation decisions compared to auto-pay users.

#(6) Business Recommendations:

- **Target Month-to-Month Extensions:** Run email and SMS promotional campaigns targeting monthly users. Offer a small discount (e.g., 10% off) or a free speed upgrade if they transition to a locked-in 1-year or 2-year contract.

- **Incentivize Automatic Billing Setup:** Introduce a one-time bill credit incentive (such as a \$5 discount) for any customer who switches their payment preference from manual electronic checks to automatic Credit Card or Bank Transfer auto-debit.
- **Implement a New User Check-In Protocol:** Establish a customer success outreach program where support agents proactively call or run diagnostic checks for new subscribers at day 30 and day 90 to resolve setup complaints or speed issues before users abandon the service.

#(7) Conclusion:

This exploratory data analysis highlights that a company's churn problem is rarely random; it follows clear operational patterns. Our calculated churn rate of 16.8% can be systematically reduced. By focusing business strategies on migrating users away from loose month-to-month terms, removing manual billing friction, and giving extra technical support to new sign-ups during their first 90 days, the company can successfully plug its revenue leaks and secure long-term subscriber growth.